

Bryce Rogers

Ph.D., M.S.

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SUMMARY

Lover of Christ and His word, aiming to work in web, product & data-based development for His glory. Ph.D. computational research scientist with 4.5 years graduate-level data science, modeling, analysis, visualization and development experience (Python & Git ecosystems). Web applications from idea to versionable app in days or weeks. 7 years vocational, graduate-level quantitative science and communication.

CORE COMPETENCIES

- Languages: *Python, AI-assisted: JavaScript, HTML, CSS*
- AI & Development Tools: IDE & Web-based *AI-assisted backend/frontend development (very competent with GitHub Copilot/Copilot Chat: agentic PRs, local and cloud agents, prompt engineering, model selection, human-guided agentic code review testing & documentation)*, VPN/SSH/SFTP, HPC Workloads, Virtual Environments, UNIX/Bash CLI, VPN, GitHub, Fork (Git)
- Modeling & Analysis: Predictive modeling (machine learning, deep learning, and statistical methods; regression, classification), custom neural networks (image & pseudo-speech classification), exploratory data analysis, cleaning & transformation (e.g. standardization), statistical inference (e.g. uncertainty estimation), visualization
- Communication: Coherent technical documentation and presentation across reports, slides, or figures; substantive experience communicating scientific or analytical concepts verbally, visually or in writing to technical and non-technical audiences
- Engineering & Solution Cycle: Learn the problem landscape, define the goal, outline steps toward the goal, implement your steps, collect your data, interpret it, learn from it

EXPERIENCE

Independent Developer | Kannapolis, NC | 2025-26 (1.25 years)

- AI-assisted web application development (*GitHub Copilot, JavaScript, HTML, CSS*)
- Created and developed a *Bible learning app "Iron: Sharpen Bible Knowledge"* (www.iron-sharpen.com).
 - Approx. 3 weeks from hand-made images and notes to a stable, standalone web version with rich UI/UX
 - Full-size/tablet design, plus mobile compatibility features (e.g., window resize, touch-to-hold priority). GitHub Pages-hosted
- Created and developed an *audio visualizer app based on human sound perception*, the "Music Mix Mapper" (<https://brcodes.github.io/music-mix-mapper/>)
 - 1 week from hand-made images and notes to a stable, standalone web version with robust full-size UI/UX
 - Flex design (full-size, tablet, & mobile targets). GitHub Pages-hosted

Graduate Research Assistant | Kevin Brown Lab, Oregon State University | 2020–25 (4.5 years)

- Developed and characterized deterministic and neural-network-based predictive coding models of word recognition and vision (*Python, Git ecosystems, hybrid/remote work*)
- First known recurrent, predictive coding neural-network classifier of over-time, speech-like input with benchmark behavioral realism in human word recognition (first-author manuscript accepted for publication in 2026, *Neural Computation*)

Graduate Research Assistant | Brian McNaughton Lab, Colorado State University | 2013–16 (2.5 years)

- Graduate-level quantitative molecular biology. Assisted in characterizing the binding interaction between a designer and oncoprotein (second-author manuscript in 2014, *Biochemistry*)

FINANCIAL AWARDS

- > \$105K earned in personal academic financial awards outside of graduate assistantships. *OSU*: CoE Dean's Award (2022), ARCS Foundation Scholar (2020), Provost Distinguished Graduate Scholar (2020); *UPS*: Trustee Scholarship (2008)

EDUCATION

Ph.D. Bioengineering (Biological Data Sciences Minor) | Oregon State University | 2020–25 | GPA: 3.95

- Computational modeling, data science & machine learning in biology
- Data science, programming and analysis-intensive courses: Programming & Data Structures, Applied Machine Learning, Command Line Data Analysis (UNIX/Bash), Intro to UNIX/LINUX, Data Science Tools & Programming, Modern Statistical Methods for Large & Complex Datasets, Foundations of Data Analytics (Python, R, Bash)

M.S. Chemistry | Colorado State University | 2013–16 | Ph.D. Candidacy | GPA: 3.52

B.S. Biochemistry, Spanish | University of Puget Sound | 2008–12 | GPA: 3.05

PUBLICATIONS

- "Extending a formal model of predictive coding to human spoken word recognition." — B.R., M.Y.L., T.H., J.S.M., and K.B. Accepted in *Neural Computation*, 2026
- "Predictive coding models in human spoken word recognition." — B.R. *OSU* 2025
- "Characterization of the Binding Interaction between the Oncoprotein Gankyrin and a Grafted S6 ATPase." — A.C., B.R., and B.M. *Biochemistry* 2014

FORMAL COMMUNICATION

- Formal poster presentations:
 - "Predictive Coding Models in Human Cognition." *ARCS Annual Scholar Event* (2022), *Oregon Bioengineering Symposium* (2022)
 - "Formal Predictive Coding Models in Human Vision and Beyond." *Oregon Bioengineering Symposium* (2021)